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Marathousa 1: a new Middle Pleistocene archaeological site from Greece

Eleni Panagopoulou, Vangelis Tourloukis, Nicholas Thompson, Athanassios Athanassiou, Georgia Tsartsidou, George E. Konidaris, Domenico Giusti, Panagiotis Karkanas & Katerina Harvati

Introduction

Lower Palaeolithic evidence in Greece is sparse and consists of unprovenanced finds or sites with material deriving from secondary contexts (Tourloukis & Karkanas 2012). The basin of Megalopolis, Greece, has long been known for its Pleistocene fossiliferous sediments (e.g. Melentis 1961). Early human activity is suggested by a hominin tooth collected as a surface find (see Harvati *et al.* 2009), as well as by observations of lithic artefacts (Darlas 2003). Nevertheless, systematic archaeological research has been lacking to date. Here, we report the first results from the excavation of 'Marathousa 1', a primary-context open-air site from Megalopolis.

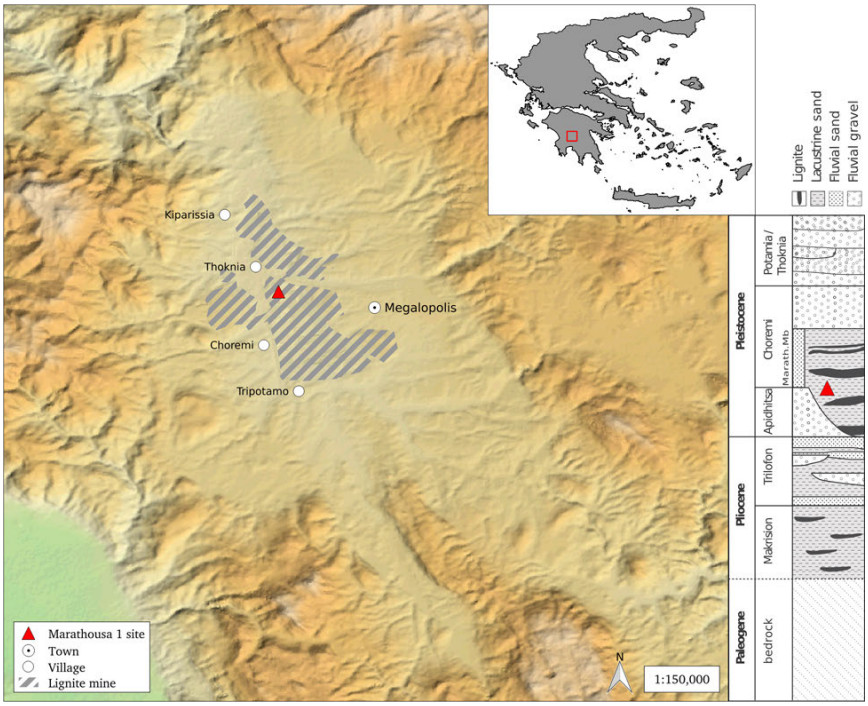

Figure 1. Map and stratigraphic column (modified after van Vugt *et al.* 2000) of the Megalopolis basin.

The geological sequence of the basin includes lacustrine and fluvial deposits that are divided into six formations; of particular interest is the Marathousa Member (Choremi Formation), which dates to the Middle Pleistocene and is composed of lacustrine clay, silt and sand beds alternating with lignite seams (Vinken 1965; van Vugt *et al.* 2000; Figure 1). These deposits represent the environment of a large lake, which mainly covered the western half of the basin (in addition to some other parts) and periodically became a shallow swamp. The detrital beds were most likely formed during cold/dry periods, and they correlate with glacials (or stadials), while the lignite beds represent interglacials (or interstadials). Since 1969, the lignite seams have been exploited via open-cast mines and numerous palaeontological localities have been exposed during mining operations.

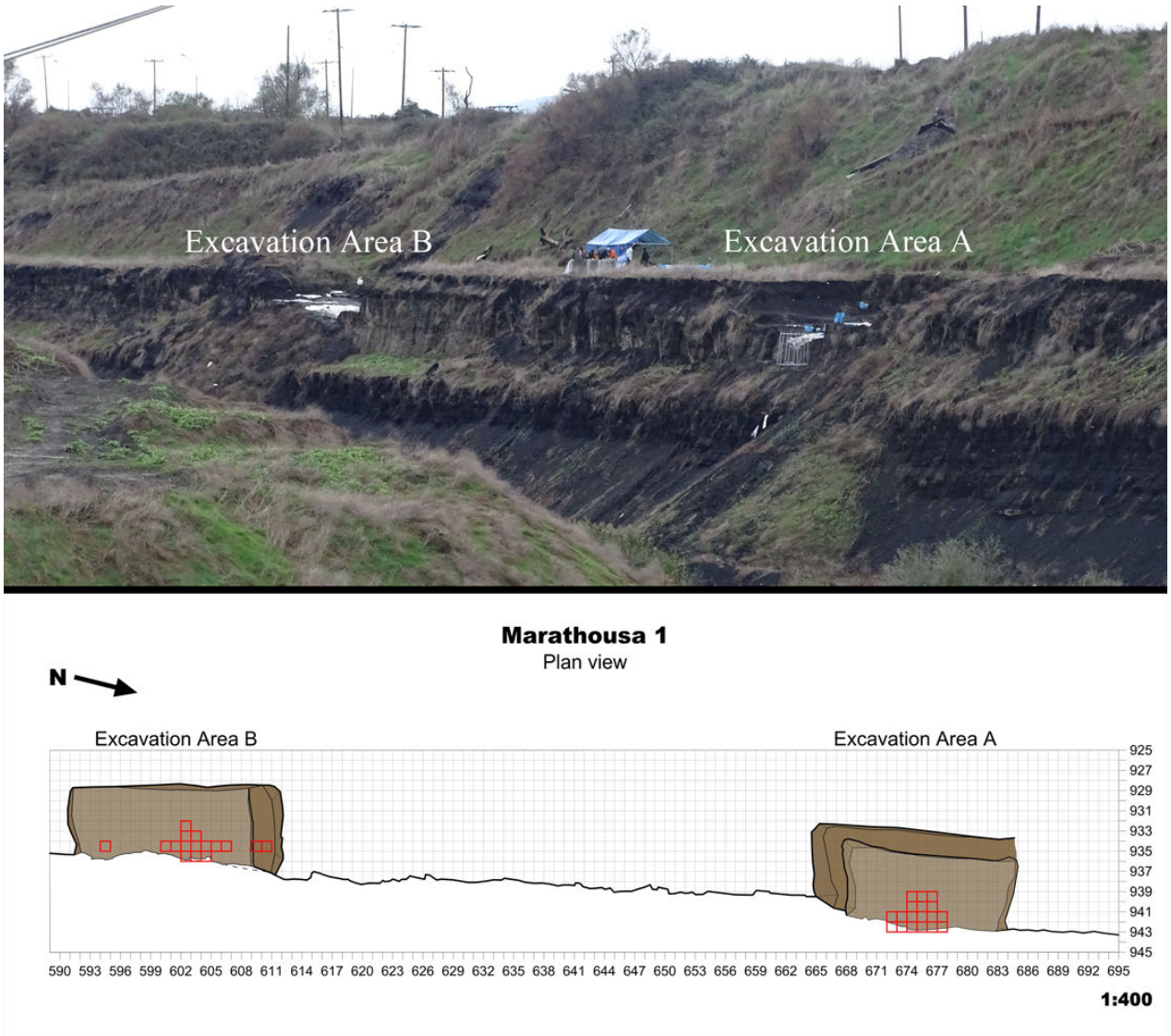

Figure 2. Above: panoramic view towards the south-west; below: plan view of the site, showing the grid and excavation areas.

The site

'Marathousa 1' was discovered in 2013 during an archaeological survey conducted by a team from the Ephoreia of Palaeoanthropology-Speleology of southern Greece and the University of Tübingen. The site was located when stratified bones and artefacts were identified in a profile of the Marathousa Member. After the initial discovery, the remainder of the 2013 field season was devoted to rescue excavations in order to collect, or to cover with plaster-jackets, material that was in danger of eroding out of the section. The results presented here refer to data collected during the salvage of archaeological work and test pitting of 2013, and also to data collected during the first season of systematic excavations in 2014. Fossil bones and lithic artefacts occur in varying densities for around 100m along the profile, but the exact spatial extent of the site is yet to be established. Excavation units of 1×1m were opened in two excavation areas, A and B (Figure 2).

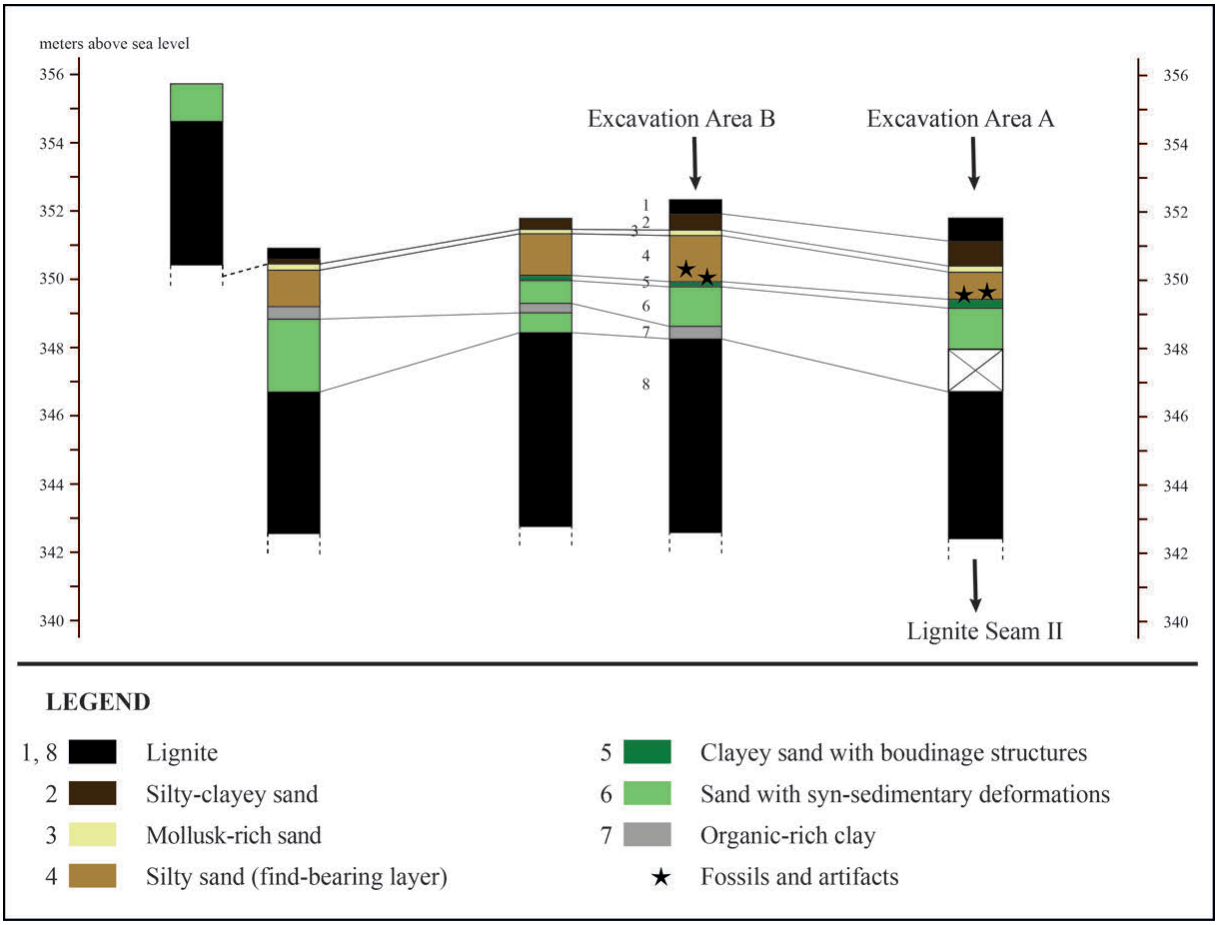

Figure 3. Stratigraphic columns showing the geological sequence of the site.

Preliminary study of the stratigraphy suggests that finds occur inside Layer 4 and at the point of contact between Layer 4 and Layer 5 (Figure 3). Layer 4 is a 0.6–1.8m-thick deposit

of dark brown silty sand, rich in organic remains and interbedded with lenses of fine sands, some of which are laminated. The underlying Layer 5 is composed of greyish clayey sand, and it is more homogeneous in structure than the previous layer. In addition to lithic artefacts, and with the exception of small (usually < 20mm) sandstone and limestone gravels that occur very sporadically, all layers in the sequence generally lack any rocks. Our working hypothesis is that the site context represents a low-energy depositional environment, such as a shallow-water swamp close to the shore of the lake. As shown in Figure 3, the find-bearing layers occur between lignite seams II and III; consequently, according to the chronostratigraphic model of van Vugt *et al*. (2000), 'Marathousa 1' dates to the middle part of the Middle Pleistocene.

The finds

In 2013, our work focused on the cranium of an elephantid. Taxonomic identification, based on the craniodental morphology, attributes it to the species *Elephas (Palaeoloxodon) antiquus*. The expansion of excavation units in 2014 revealed numerous, well-preserved elephant skeletal elements in the same layer, including a femur, ribs, vertebrae, an astragalus, a patella and tusk fragments (Figure 4). Their stratigraphic and close spatial association with the cranium suggests that they belong to the same individual. Other faunal material from the site includes teeth, mandibular and postcranial remains of

Figure 4. Excavation area A: panoramic view of the elephant bones. The plaster-jacket in the background covers the cranium. Scale: 50cm; excavation units: 1×1m.

mammals, such as cervids and bovids, as well as micromammals, turtles and birds. Additional elephant elements, including the proximal end of a tibia, were found *c.* 50m from the main elephant-fossil accumulation (Figure 5). If they prove to belong to the same individual, they could indicate either bone dispersal by natural processes or, more likely, transport by hominins. The direct contextual association of artefacts and fossils (notably, elephant bones) encourages further research on the involvement of hominins in the exploitation of these animal carcasses.

Figure 5. Excavation area B, showing the elephant tibia fragment in situ.
The red circle marks the position of a radiolarite flake.

The lithic assemblage (n=122) is composed of flakes and flake fragments, core fragments and chunks (Figure 6). Retouched tools are rare. The main raw material is red and brown radiolarite, followed by grey and black flint, limestone and quartz. Platforms are mainly plain, cortical or dihedral, and they indicate hard hammer percussion. Technological traits suggest a relatively simple operational sequence, probably aiming at the production of flake blanks. Flakes might have been used directly for cutting, without further modification. Lacking bifacial elements, the lithic assemblage from Marathousa could be ascribed provisionally to a 'core-and-flake' techno-complex, although the sample is currently limited. Notably, many red radiolarite artefacts seem to derive from the same working

piece of raw material, perhaps indicating an *in situ* and short-lived knapping episode. All artefacts are exceptionally well-preserved, with no evidence of rolling, supporting our hypothesis of low-energy depositional agents and limited displacement from the original discard locations.

Figure 6. Lithic artefacts from the 2013 field season.

Conclusions

'Marathousa 1' is an open-air site, situated on what would have been the shores of a palaeo-lake, where elephant and other faunal remains are found in association with stone tools. The site probably testifies to hominin activities that pertain to the exploitation of animal resources. The finds occur in a very fine-grained geological matrix that has fostered exceptional preservation. This is the first Middle Pleistocene archaeological site ever to be excavated in mainland Greece and, moreover, the first Greek, open-air Lower Palaeolithic site with both lithic and faunal remains to be examined by systematic excavation. Consequently, 'Marathousa 1' offers a unique opportunity to study early human behaviour in a region that has long remained virtually unexplored.

Acknowledgements

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